

How Ricardian Are We?

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“The people who pay the taxes never so estimate them, and therefore do not manage their private affairs accordingly.”

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 - the aggregate **MPC out of debt-financed transfers**?

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Object of Interest

θ = share of future taxes households save to pay for

Why Micro Evidence Is Not Enough

What Existing Micro Evidence Tells Us

How much consumption rises when a household receives more current resources.

This identifies the usual marginal propensity to consume (MPC).

What The Ricardian Question Needs

How much consumption rises after a debt-financed transfer or tax cut.

That is the MPC out of debt-financed transfers (MPC-DFT).

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That is the MPC out of debt-financed transfers (MPC-DFT).

- The missing intercept problem: micro evidence identifies MPC, not the MPC-DFT
- We take a semi-structural route: estimate a consumption function linking consumption to taxes, debt, wealth, expected income, and bond prices.

Why A Consumption Function?

Modest Ingredients

- household budget constraint
- Euler equation for bond holdings (with a non-Ricardian wedge)
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- Many potential sources of non-Ricardian behavior map to the same consumption function
 - Clearest case: households have non-rational expectations over future taxes (internalize share θ only)
 - Consumption function coefficients identify θ (but not with OLS...)

Our Approach

1. Derive the behavioral consumption function from budget constraints + Euler equation
2. Estimate with **B-HIVE** the **B**ayesian **H**ybrid **IV** **E**stimator, using macro structural shocks as external instruments to address endogeneity
 - Consumption function has many endogenous regressors
 - Instruments cover different samples, state space methods needed
 - Priors help with weak identification, theoretical constraints
3. Feed the estimate into a general equilibrium model to quantify the consequences of public borrowing

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Goal

Estimate non-Ricardian behavior without taking a stance on a fully specified equilibrium model

Consumption Function

$$c_t = \phi_0 + \phi_n n_{t-1} + \phi_b b_{t-1} + \phi_\tau \tau_t + \tilde{\phi}_y v_t^y + \tilde{\phi}_g v_t^g + \tilde{\phi}_q v_t^q + \zeta_t$$

- n_{t-1} : net worth, b_{t-1} : debt, τ_t : taxes.
- v_t^y, v_t^g, v_t^q : sums of discounted expected values of income, spending, and bond prices.
- $\phi_n = (1 - \beta)$: the MPC
- $\phi_\tau = -(1 - \beta)(1 - \theta)$: the MPC-DFT
- $\theta = 1$: Ricardian. $\theta < 1$: taxes/transfers move consumption

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Why Micro Evidence Is Not Enough (Revisited)

Cross-sectional designs vary who gets the transfer, not the common **future tax burden**; that is absorbed in the time fixed effect, so they do not recover ϕ_τ .

What Can Generate This Consumption Function?

In The Paper, We Derive This Object Under Three Sets Of Assumptions

$$c_t = \dots - (1 - \beta)(1 - \theta)\tau_t + \dots$$

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$$\tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^\tau] = \theta \mathbb{E}_t[v_{t+1}^\tau]$$

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$$v_t^\tau = \rho_\tau v_{t-1}^\tau + \varepsilon_t^\tau, \quad \theta = \omega \frac{1 - \beta\rho_\tau}{1 - \beta\omega\rho_\tau}$$

then the consumption function is **isomorphic** to the behavioral one.

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3. **Two-agent / hand-to-mouth economy**: with a share $1 - \lambda$ of constrained households,

$$c_t^C = y_t^C - \tau_t, \quad \theta = 1 - \frac{1 - \lambda}{1 - \beta}$$

Why Some Obvious Approaches Miss ϕ_τ

1. **Full-equation OLS:** the demand shock ζ_t is omitted, but in general equilibrium it also moves observed RHS variables. So regressors are correlated with the error term, and OLS cannot isolate ϕ_τ . (Feldstein, 1982; Cardia, 1997)

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2. **Lagged aggregates as IV:** if ζ_t is persistent, lagged consumption, wealth, or asset prices inherit past ζ_t and remain correlated with today's residual. So the exclusion restriction fails. (Hayashi, 1982; Feldstein, 1982; Seater and Mariano, 1985)

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3. **Regressing consumption on exogenous tax shocks:** this estimates the total equilibrium response to a tax change, not the direct tax coefficient ϕ_{τ} . Tax shocks also move income, returns, and asset prices. (Mertens and Ravn, 2012; Leeper, Richter, and Walker, 2012)

Takeaway

None of these approaches isolate the direct tax coefficient ϕ_{τ} .

Barnichon-Mesters: Structural Shocks as IVs

- Key idea: use externally identified macro shocks to estimate one structural equation. Barnichon and Mesters (2020, QJE)
- For the consumption function, the instruments must move the endogenous RHS variables while remaining excluded from the residual ζ_t (EE wedge)
- Intuition: productivity, monetary, fiscal, oil, etc. shocks all shift expected income, returns, wealth, taxes, or bond prices

Identification Logic

Use shocks that are excluded from the consumption equation but informative about its endogenous regressors.

B-HIVE Estimation

$$c_t = \phi' X_t + \zeta_t$$

$$X_t \equiv [1, n_{t-1}, b_{t-1}, \tau_t, \dots]'$$

$$X_t = \mu_X + A_1 X_{t-1} + \dots + A_p X_{t-p} + G \varepsilon_t$$

$$w_t = \mu_w + M_X X_{t-1} + M \varepsilon_t + \eta_t$$

- estimate one structural equation
- use the VAR in X_t to compute expected present values
- observe noisy shock instruments w_t
- Bayesian state space handles missing data, weak-IV concerns

State-Space Representation

Discounted Sums From The VAR

$$v_t^j = s_j'(I_{mp} - \beta_j F)^{-1} \bar{X}_t$$

where F is the companion-form transition matrix and $\bar{X}_t$ stacks current and lagged values of X_t .

- The present values on the RHS of the consumption equation are functions of the VAR coefficients.
- That creates cross-equation restrictions: the structural equation and state dynamics are estimated jointly.
- The state-space likelihood handles missing observations and different sample sizes naturally.

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Benefits of Our Approach

- model-consistent present values
- noisy external instruments
- Bayesian regularization under weak IV
- Standard Bayesian VAR toolkit available

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Extensions

The same framework can be extended to **non-Gaussianity**, **stochastic volatility**, etc.

Data Part 1: Macro Time Series

Observables

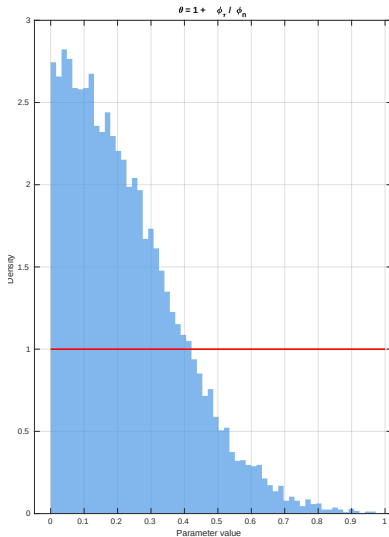
Quarterly U.S. data, 1969:Q1–2019:Q4.

- personal consumption
- personal taxes net of transfers
- personal income net of transfers
- household net worth
- market value of federal debt
- government expenditures
- 3-month Treasury bill yield
- SPF inflation forecasts
- real consol value (Aruoba term structure)
- survey income forecasts
- additional FRED-QD macro factors in X_t

Data Part 2: Macro Shocks (IVs)

Category	Examples used in the paper
Monetary policy	Jarocinski-Karadi, Miranda-Agrippino-Ricco, Bauer-Swanson, Aruoba-Drechsel
Government spending	Fisher-Peters, Ramey military news, Romer-Romer transfers, Fieldhouse-Mertens-Ravn housing, Fieldhouse-Mertens R&D
Tax / borrowing	Leeper-Walker-Yang, Mertens-Ravn, Lieb et al., Phillot
Technology	Fernald TFP, Miranda-Agrippino et al. patent news
Oil	Kilian, Kanzig, Baumeister-Hamilton
Other	Piffer-Podstawski uncertainty, Kim et al. severe weather, Chahrour-Jurado noise, Adams-Barrett inflation-expectations

Main Result I: Households Are Far From Ricardian



Behavioral Attenuation

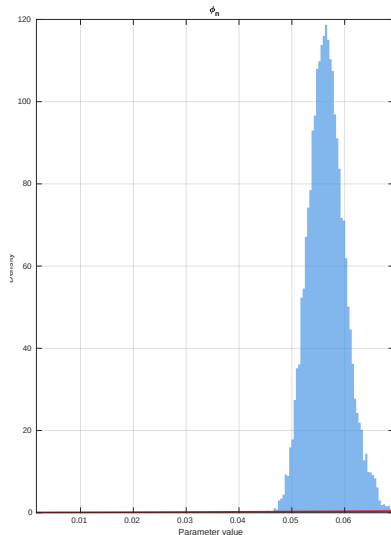
$$\theta = 0.19$$

90% credible interval:

$$[0.02, 0.54]$$

- $\theta = 1$ is the Ricardian benchmark.
- The posterior mass lies far below that benchmark.
- Baseline interpretation: households internalize only about **one-fifth** of future taxes.

Main Result II: The Generic MPC Is Small But Precisely Estimated



Quarterly MPC

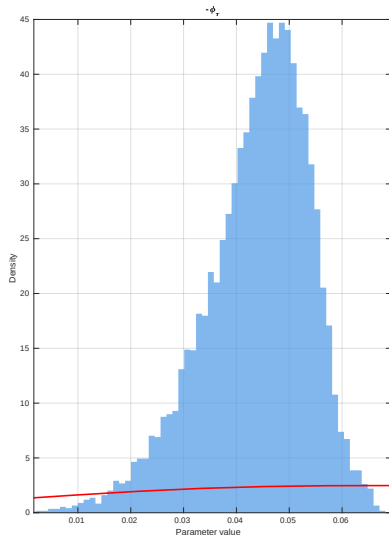
$$\text{MPC} = 0.056$$

90% credible interval:

$$[0.051, 0.063]$$

- The aggregate marginal propensity to consume out of net worth
- Prior is flat, posterior is tight
- Empirical estimates for MPCs out of illiquid wealth: 3%-7%

Main Result III: Debt-Financed Transfers Move Consumption



MPC Out Of Debt-Financed Transfers

$$\text{MPC-DFT} = 0.045$$

90% CI: [0.025, 0.058]

$$\text{MPC-DFT} = -\phi_\tau = (1 - \beta)(1 - \theta)$$

- Using $\phi_n = \text{MPC}$, we infer $\theta = 1 + \phi_\tau / \phi_n$
- Debt-financed transfers clearly **increase consumption**

Robustness: The Answer Stays Non-Ricardian

Selected Robustness Checks

Specification	θ	MPC-DFT
Baseline	0.194	0.045
Non-durable consumption	0.130	0.050
Survey-based forecasts	0.231	0.007
Non-separable utility	0.096	0.059
HtM income shares	0.123	0.066
Six selected instruments	0.174	0.027

- Each check targets a specific concern: richer expectations, labor in the SDF, HtM households, consumption measurement, or a leaner instrument set. [Full table](#)

From The Estimate To General Equilibrium

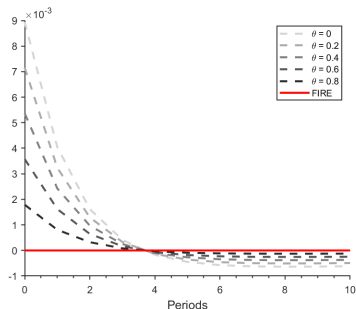
- Embed the estimated consumption function in a simple closed-economy RBC model with capital.
- A debt-financed tax cut makes households feel richer when $\theta < 1$.
- So consumption rises on impact, saving falls, and investment is crowded out.
- This is the equilibrium implication the reduced-form estimate is pointing to.

Mechanism

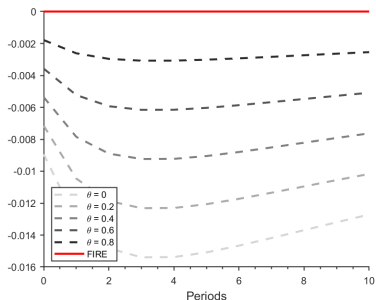
lower taxes today $\Rightarrow$ higher perceived wealth $\Rightarrow$ higher $c_t \Rightarrow$ lower $i_t \Rightarrow$ lower k_{t+1}

Equilibrium Implication: Government Debt Crowds Out Capital

Consumption response



Capital response



- Negative tax shock = debt-financed transfer.
- The smaller is θ , the stronger is the rise in consumption and the fall in capital.
- Quantitatively, our estimate implies substantial crowding out of private investment.

Takeaways

- The economic question is old, but the object we need is clear: the aggregate marginal propensity to consume out of debt-financed transfers
- Micro data are not enough: we need aggregate time series!
- Estimating the consumption function is a semi-structural strategy
- Barnichon-Mesters' insight: use other structural shocks to identify the equation
- Empirically, households are far from Ricardian: baseline $\theta \approx 0.19$.
- Combination of discounting wedges and non-rational expectations can rationalize
- In GE, non-Ricardianism implies that government borrowing crowds out capital.

Method

Monte Carlo

Mechanisms

Appendix: Full Consumption Function

$$c_t = (1 - \beta) \left(n_{t-1} - \theta b_{t-1} + \tilde{v}_t^y - (1 - \theta) \tau_t - \theta v_t^g + \theta \bar{B} v_t^q \right) + \left(\frac{\bar{c}}{\gamma} - (1 - \beta) \bar{B} \right) \tilde{v}_t^q + \zeta_t$$

- $(1 - \theta) \tau_t$ is the direct non-Ricardian tax term.
- Behavioral expectations motivate this mapping directly.
- Discounting wedges generate an **isomorphic** consumption function in the paper.
- HtM or liquidity-friction models imply a very similar aggregate equation in which current taxes still matter.

Appendix: B-HIVE In One Slide

$$c_t = \phi_0 + \phi_n n_{t-1} + \phi_b b_{t-1} + \phi_\tau \tau_t + \sum_{j \in \{y, g, q\}} \tilde{\phi}_j v_t^j + \zeta_t$$

$$X_t = \mu_X + A_1 X_{t-1} + \cdots + A_p X_{t-p} + G \varepsilon_t$$

$$w_t = \mu_w + M_X X_{t-1} + M \varepsilon_t + \eta_t$$

- The VAR for X_t pins down the discounted expected-value terms.
- The measurement equation links observed shock series w_t to latent structural shocks ε_t .
- Bayesian inference helps with weak instruments and lets us impose economically natural bounds such as $0 \leq \theta \leq 1$.
- State-space methods naturally handle missing instruments and mixed sample coverage.

Appendix: Full Robustness Table

Specification	θ	MPC	MPC-DFT
Baseline	0.194	0.056	0.045
Non-durable consumption	0.130	0.058	0.050
Survey-based forecasts	0.231	0.009	0.007
Non-separable utility	0.096	0.065	0.059
Variable distortionary taxes	0.597	0.058	0.023
HtM income shares	0.123	0.076	0.066
Including SVAR instruments	0.123	0.063	0.055
Six selected instruments	0.174	0.033	0.027

[Back](#)

Appendix: Non-Rational Expectations Vs. Discounting

How The Decomposition Works

Under additional assumptions on tax dynamics, β

$$\phi_n = 1 - \beta\omega, \quad -\phi_\tau = (1 - \beta\omega)(1 - \theta_{cd})$$

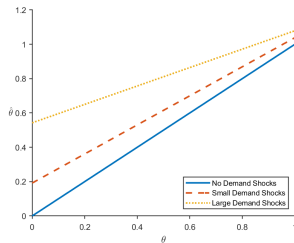
so two reduced-form moments pin down the discounting wedge ω and expectations wedge λ

Mechanism	λ	ω
Naive cognitive discounting	0.20	
Sophisticated cognitive discounting	0.90	
Discounting wedge + rational		0.90
Discounting wedge + naive C.D.	0.59	0.95
Discounting wedge + sophisticated C.D.	0.95	0.95

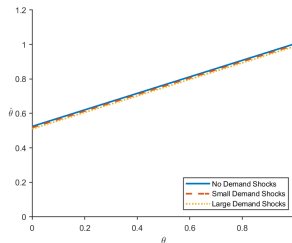
- Matching both MPC and MPC-DFT requires both $\lambda < 1$, $\omega < 1$ [Back](#)

Appendix: Monte Carlo I

OLS



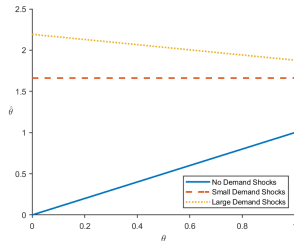
Tax shocks only



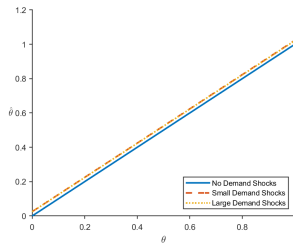
- OLS is biased when demand shocks move both consumption and endogenous regressors.
- “Just use tax shocks” is also biased because those shocks operate through other equilibrium margins.

Appendix: Monte Carlo II

Lagged variables as IVs



Structural-shock IVs



- Lagged aggregates fail once omitted shocks are persistent.
- Structural-shock IVs recover the correct coefficient.